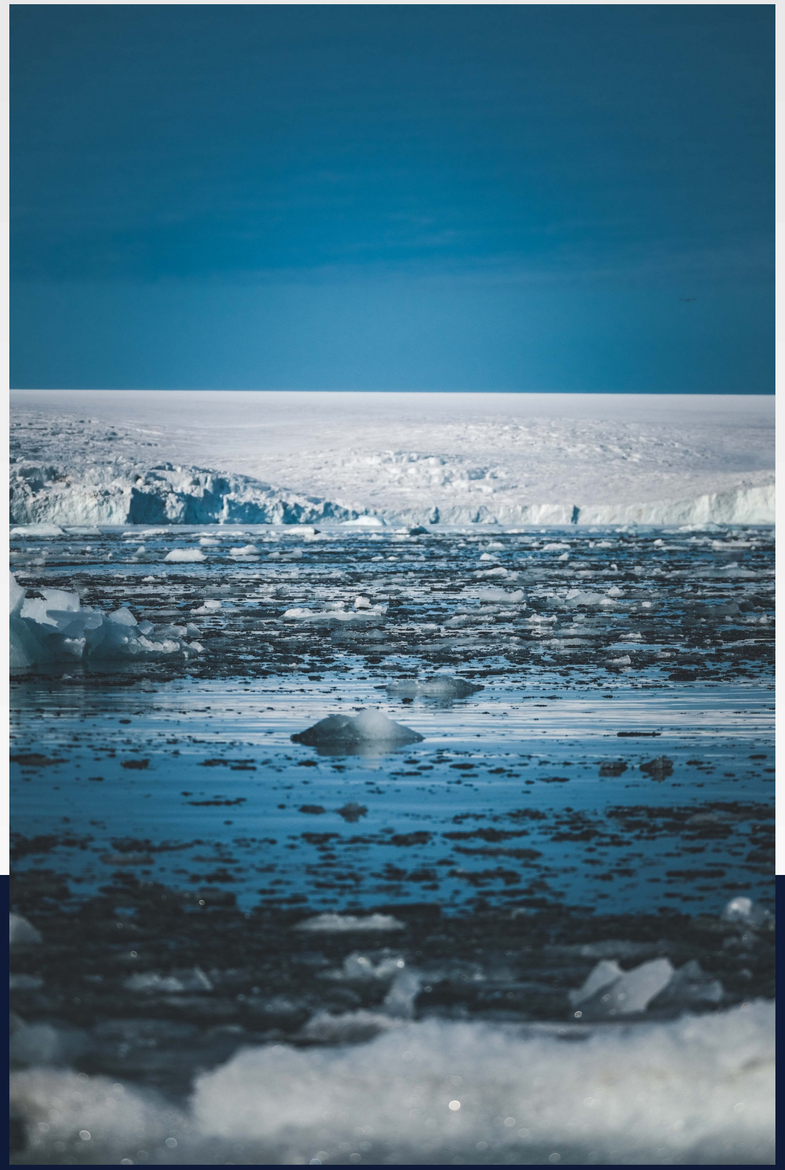


EXECUTIVE SUMMARY

Royal Greenland Digital Transformation

SAP Customer Analysis
& Architecture Assessment

Aditya Arya – Technical Consultant
October 2026

STRATEGIC CONTEXT & OBJECTIVES

Royal Greenland Customer Context & Transformation Drivers

Global Seafood exporter coordinating artisanal catch procurement across remote Arctic coastal sites. Primary focus: Eradicate paper logs, automate GFLK/MSC quota compliance and accelerate fisher settlements.

70K+

Digital Purchase Orders

Completely eliminated physical paper logs across remote coastal receiving facilities.

200+

Active Fishers Onboarded

High voluntary mobile adoption across independent artisanal fleets with instant receipt trust.

SAP End-to-End Solution Architecture

Hybrid cloud architecture integrating offline mobile clients at sea with central SAP ERP and HANA data layers

FISHERMAN APP

IOS Native (BTP SDK)

- Offline catch registration at sea.
- Local SQLite caching & auto-sync.
- Digital PO receipt verification.

LANDING STATION APP

SAP Fiori / SAPUI5

- Harbor tablet & web interface.
- Integrates dockside scale weights.
- Quality grading & batch approvals.

PROCUREMENT PORTAL

SAP web portal / ERP

- Real-time Quota tracking.
- Pricing rules & export clearance.
- Automated settlement processing.

SAP BTP Foundation: Application Development (iOS SDK & Fiori) • Integration Suite (API Sync to Core) • Data & Analytics (Quota Tracking) • Process Automation (Digital POs)

TECHNICAL ARCHITECTURE | CLOUD & EDGE FOUNDATION

System Design & Development Considerations

Resilience & Offline-First

Local SQLite caching on mobile clients to handle intermittent Arctic maritime networks: automated background sync upon docking or signal restoration.

Polyglot Development Stacks

Purpose-built technologies: Native Swift (iOS SDK) at sea, responsive JavaScript (SAPUI5) at landing docks, and ABAP CDS views for core ERP transactions.

Hybrid Cloud Computing

Combines rapid, scalable micro-services on SAP BTP with secure enterprise financial records and purchase order processing in core SAP ERP.

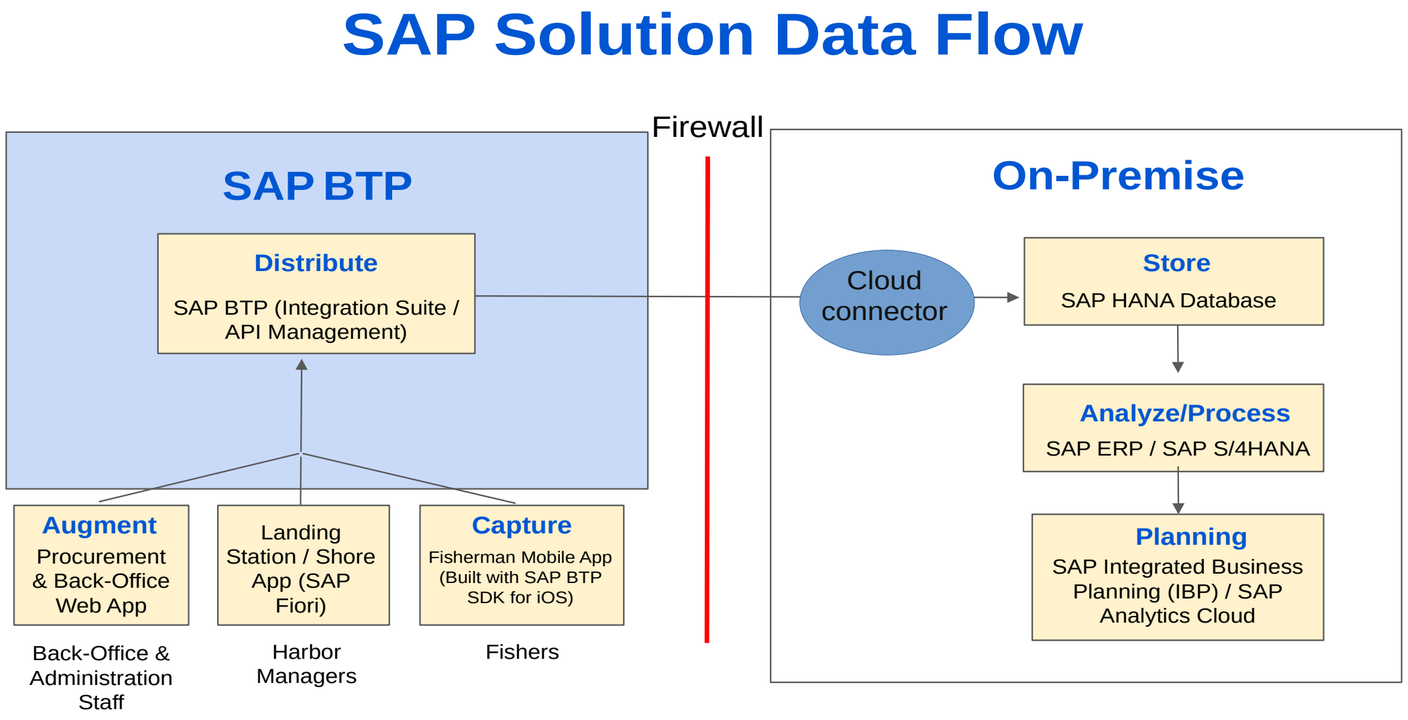
Enterprise Security & Governance

OAuth 2.0 authentication, granular Role-Based Access Control (RBAC) and end-to-end data encryption both in transit and at rest.

DATA ARCHITECTURE

Solution Data Flow

End-to-End Information Life-cycle



Capture:

Fishers log catch data offline at sea via native iOS app.

Distribute:

Secure API routing & event sync via SAP BTP Integration Suite.

Augment:

Harbor stations verify weights & grades via SAP Fiori.

Store:

High-performance persistence in SAP HANA database.

Process & Plan:

Automated digital PO settlement in SAP ERP; demand planning in SAP IBP.

THE INTELLIGENT SUSTAINABLE ENTERPRISE

Quadruple Bottom Line Value Realization

People

Fisher Empowerment & Safety

Transparent digital pricing, immediate receipt verification and eradication of tedious manual paperwork in sub-zero Arctic conditions.

Planet

Ecological Stewardship

Real-time catch registration against strict fisheries quotas: protection of Arctic marine ecosystems and retention of MSC (Marine Stewardship Council) certification.

Profit

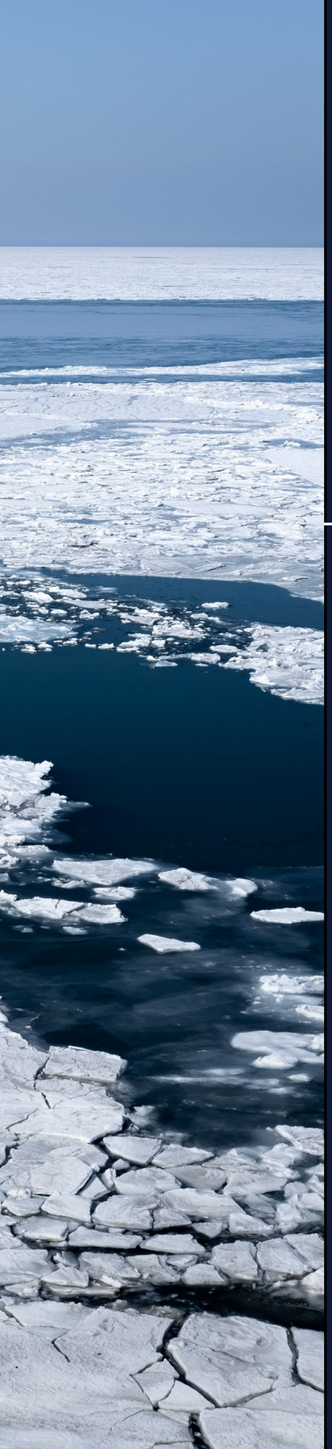
Operational & Commercial Resilience

Rapid generation of 70k+ digital purchase orders, zero billing discrepancy write-offs and optimized cold-chain factory throughput.

Purpose

Cultural & Regional Preservation

Sustaining traditional artisanal Greenlandic fishing communities by integrating them directly into the modern global seafood supply chain.



FUTURE INNOVATION HORIZONS | CONTINUOUS ENTERPRISE EVOLUTION

Next Steps & Strategic Technology Roadmap

Horizon 1

(Near-Term / Immediate Road-map):

E-Commerce & Gear-Procurement

- Deploy SAP Commerce Cloud to allow fishers to purchase gears, nets, and bait directly via mobile.
- Automate settlements by offsetting gear purchases directly against catch payout earnings.

Horizon 2

(Medium – Term / Safety & Operations):

IoT Telemetry & Vessel Tracking

- Integrate vessel IoT sensors and GPS telemetry with SAP BTP for live location tracking in open Arctic seas.
- Provide automated weather alerts and emergency SOS beacons directly to coastal search and rescue

Horizon 3

(Long-Term / Economic Empowerment):

Fin-tech & Predictive Analytics

- Embed predictive catch forecasting and micro-financing / savings tools inside the Fisherman App.
- Provide seasonal cash-flow management tools tailored to Arctic seasonal fishing cycles